

Exercise (1)

(1) Find the value of x, y if: $\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 2 & x+y \\ x-y & 5 \end{bmatrix}$

(2) Find the value of x, y, z, w if :

i) $3 \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} x & 6 \\ -1 & 2w \end{bmatrix} + \begin{bmatrix} 4 & x+y \\ x+y & 3 \end{bmatrix}$

(3) Consider the following matrices,

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 4 & 1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 0 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad E = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Compute the following (**if possible**):

i) $2A - B$, $2B - A$, $3C + B$.

ii) AC , CA , AB , BD , CD , DB , A^2 , C^2 , D^2 , E^2 , $BC - 2D$, $CB + E$

iii) $\text{Trace}(2D)$, $\text{Trace}(D^2)$, $\text{Trace}(CB)$, $\text{Trace}(2D + D^T)$, $\text{Trace}(E)$, $\text{Trace}(E^2)$

(4) If $A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 1 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -2 \\ 0 & 3 \\ 2 & 2 \end{bmatrix}$. Find the matrix X satisfies; $2A - 3B^T + 2X = O$

(5) If $A = \begin{bmatrix} 1 & y & 2 \\ 3 & 0 & z \end{bmatrix}$, $B = \begin{bmatrix} 1 & x \\ 3 & 0 \\ 1 & w \end{bmatrix}$. Find the value of x, y, z, w given that, $(AB)^T = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}$.

(6) Find the matrix A satisfies: $\left(\frac{1}{3}A^T + I\right)^T = \begin{bmatrix} -3 & 2 \\ -1 & 4 \end{bmatrix}$.

(7) If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 3 & 3 \\ -1 & 0 & 1 \\ -4 & -4 & -3 \end{bmatrix}$. Find A^{-1} and B^{-1} .

(8) If A, B , and C are 3×3 matrices such that $|3C| = 81$, $\text{trace}(C) = 5$, $(B^T - 3I)^T = 15A^{-1}$

and $A = \begin{bmatrix} 3 & 1 & 0 \\ 2 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. Find: i) The value of $\det(C^{-1})$ ii) The matrix B .

iii) The value of $|ABC|$ iv) $\text{trace}(A + 2B + C)$

(9) If $A = \begin{bmatrix} 2 & -3 \\ -1 & 1 \end{bmatrix}$ and $(AB)^{-1} = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$. Find the matrix B .

(10) If the matrix A satisfies the equation, $A^2 + 2A - I = 0$, prove that A is invertible and find A^{-1}

(25) Show that, each one of the following matrices is **orthogonal**,

$$A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix}, \quad C = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad D = \frac{1}{3} \begin{bmatrix} 2 & -2 & 1 \\ 1 & 2 & 2 \\ 2 & 1 & -2 \end{bmatrix}$$

(26) If A and B are two orthogonal matrices, then show that AB is orthogonal matrix.

(27) Find all matrices $B_{2 \times 2}$ that **commute with** $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

(28) If $A = \begin{bmatrix} x & y \\ 0 & 0 \end{bmatrix}$, find the value of x and y such that, the matrix A commutes with $B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

(29) If $A = \begin{bmatrix} 3 & -5 \\ 1 & -3 \end{bmatrix}$, evaluate, $A^6 - 4A^4 + A^2 - 2I$

(30) Show that the matrix, $A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ is **periodic** matrix and then evaluate: $A^2 + 2A - I$.

(31) Show that the matrix, $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is **periodic** matrix and then evaluate; $(A^5)^T - (A^T)^3 + I$

(32) If $A = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$. Find A^2 and A^3 and then evaluate; $3(A^5)^T - 4(A^4)^T + 5I$

(33) If $B = \begin{bmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 2 & -1 & 0 \end{bmatrix}$. Find A^2 and A^3 and then evaluate; $3(A^7)^T + 5(A^5)^T + 2I$

(34) Use table (1) to decode the message,

18 27 51 81 37 58 60 100 18 27 85 137 59 93 51 79,

which was encoded with matrix $A = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}$.

— = 0	I = 9	R = 18
A = 1	J = 10	S = 19
B = 2	K = 11	T = 20
C = 3	L = 12	U = 21
D = 4	M = 13	V = 22
E = 5	N = 14	W = 23
F = 6	O = 15	X = 24
G = 7	P = 16	Y = 25
H = 8	Q = 17	Z = 26

(35) Find the original message that has been written in a matrix form,

$B = \begin{pmatrix} 19 & 19 & 5 \\ 25 & 20 & 13 \end{pmatrix}$ and then encode this message using the

encoding matrix, $A = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$.